The OSI Model

7-Layer Network Architecture

Mapping network functions to their security implications

7 APP	6 PRES	5 SESS	
4 TRANS	3 NET	2 DATA	1 PHYS

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What is the OSI Model?

DEFINITION

Open Systems Interconnection — a conceptual framework that standardizes network communication into 7 distinct layers.

Developed by ISO in 1984.

WHY IT EXISTS

- Vendor-neutral standard
- Diagnose network issues
- Map attacks to exact layers
- Guide security architecture

OSI vs TCP/IP

7 Layers
→ 4 Layers
Theoretical
→ Practical
Diagnostic tool
→ Internet standard

KEY INSIGHT

Security professionals use OSI to pinpoint exactly where an attack executes.

How to remember from layer 7-1?

7	Application Layer
6	Presentation Layer
5	Session Layer
4	Transport Layer
3	Network Layer
2	Data Link Layer
1	Physical Layer

All

People

Should

Try

New

Domino's

Pizza

How to remember from layer 1-7?

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Please
Do
Not
Throw
Sausage
Pizza
Away

SLIDE 05

Layer 1 —

Physical

L1

LAYER

RESPONSIBILITY

Transmits raw bit streams over the physical medium — cables, fiber optics, hubs, radio frequencies, and electrical signals.

APPSEC CONTEXT

Attacks involve physical security breaches: hardware wiretapping, rogue device insertion (keystroke loggers, Rubber Ducky), and physical denial of service.

PRIMARY THREAT

Physical Tapping / Rogue HW

RELEVANT TOOLS

- Rubber Ducky
- Proxmark
- LAN Turtle

SLIDE 06

Layer 2 —

Data Link

L2

LAYER

RESPONSIBILITY

Handles node-to-node data transfer within a local network using physical (MAC) addressing. Managed by switches and NICs.

APPSEC CONTEXT

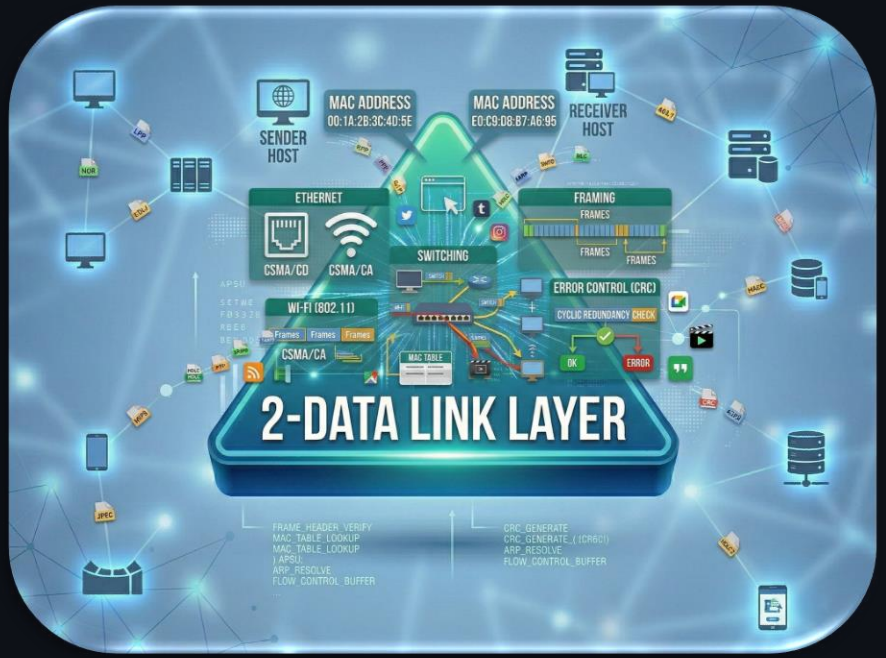
Local Area Network (LAN) attacks originate here. ARP Spoofing is the most prominent attack, allowing interception of traffic between devices on the same network.

PRIMARY THREAT

ARP Spoofing / MITM

RELEVANT TOOLS

- Ettercap
- arpspoof
- Bettercap



SLIDE 07

Layer 3 —

Network

L3

LAYER

RESPONSIBILITY

Manages logical addressing (IP addresses) and routing — determining the path data takes across interconnected networks.

APPSEC CONTEXT

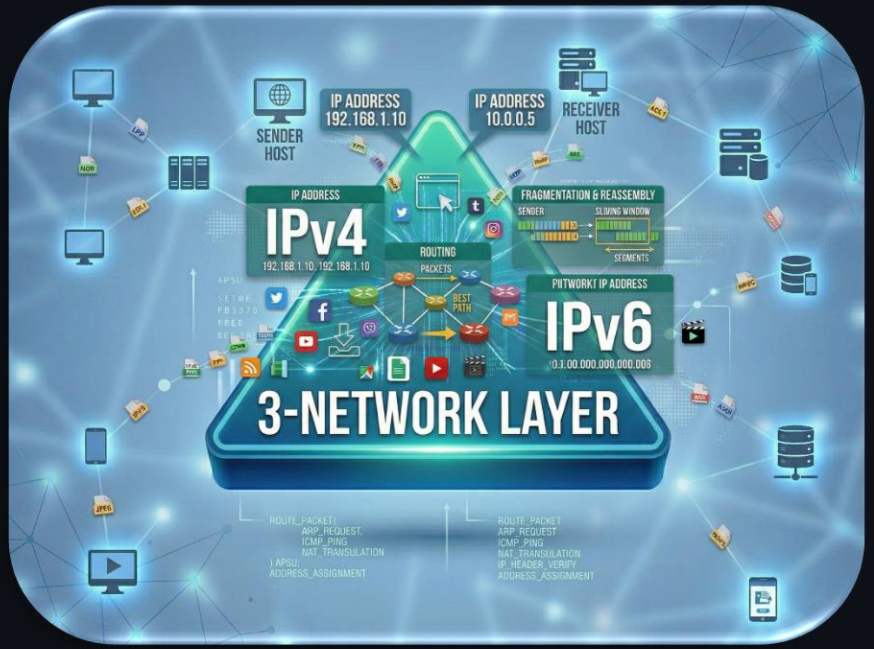
Network boundary testing occurs here. Assessments include firewall rule analysis, IP Spoofing attempts, and routing protocol attack testing.

PRIMARY THREAT

IP Spoofing / Firewall Bypass

RELEVANT TOOLS

- traceroute
- hping3
- Scapy



SLIDE 08

Layer 4 —

Transport

L4

LAYER

RESPONSIBILITY

Provides end-to-end data delivery using TCP (reliable, ordered) and UDP (fast, connectionless). Relies on port numbers.

APPSEC CONTEXT

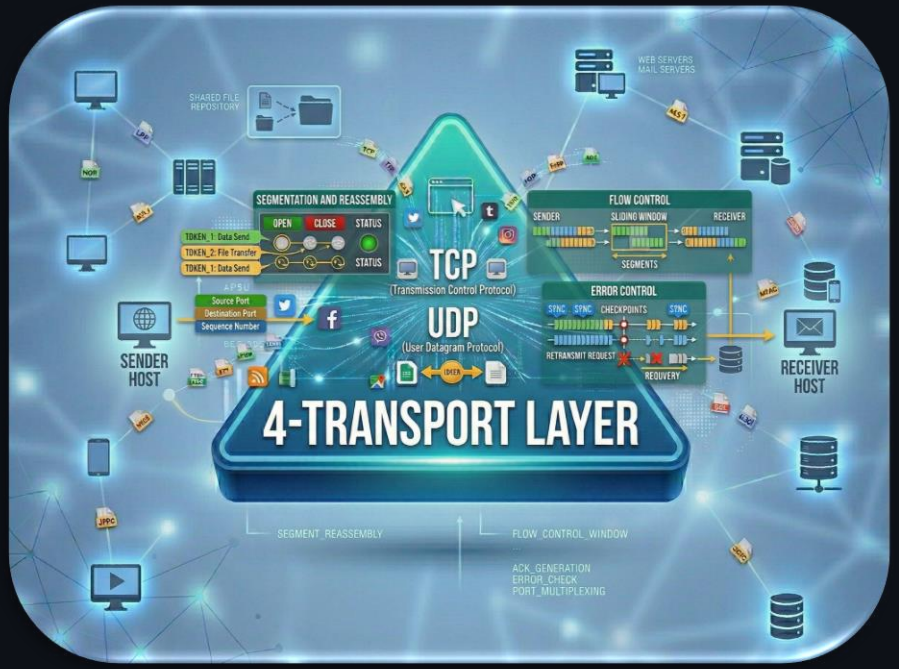
Primary layer for reconnaissance operations. Port Scanning and service enumeration are conducted here to map open attack vectors before exploitation.

PRIMARY THREAT

Port Scanning / Enumeration

RELEVANT TOOLS

- Nmap
- Netcat
- Masscan



SLIDE 09

Layer 5 —

Session

L5

LAYER

RESPONSIBILITY

Establishes, manages, and terminates communication sessions between local and remote applications.

APPSEC CONTEXT

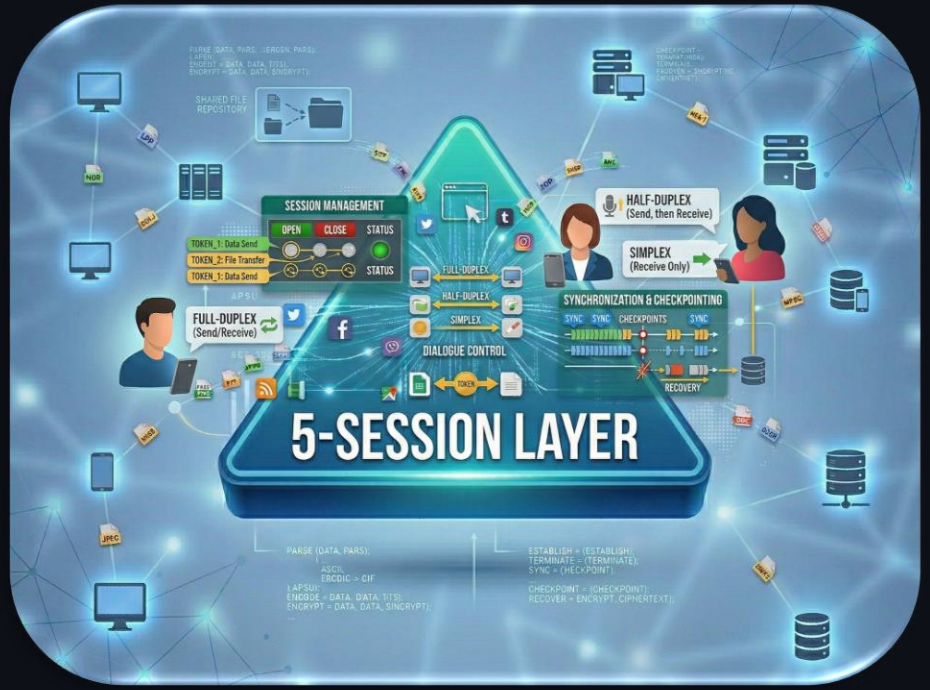
Session Hijacking and Session Fixation attacks occur at this layer. Attackers steal or forge valid session tokens to impersonate authenticated users.

PRIMARY THREAT

Session Hijacking / Fixation

RELEVANT TOOLS

- Burp Suite
- Cookie Manager
- Wireshark



SLIDE 10

Layer 6 —

Presentation

L6

LAYER

RESPONSIBILITY

Responsible for data translation, encryption, compression, and formatting before delivery to the Application layer.

APPSEC CONTEXT

SSL/TLS vulnerabilities are assessed here. Attacks like SSL Stripping, certificate validation failures, and weak cipher suites are identified at this layer.

PRIMARY THREAT

SSL Strip / Weak Ciphers

RELEVANT TOOLS

- Wireshark
- SSLScan
- TestSSL.sh

SLIDE 11

Layer 7 —

Application

L7

LAYER

RESPONSIBILITY

The human-computer interaction layer. Handles HTTP, FTP, DNS, SMTP —all protocols users directly interact with.

APPSEC CONTEXT

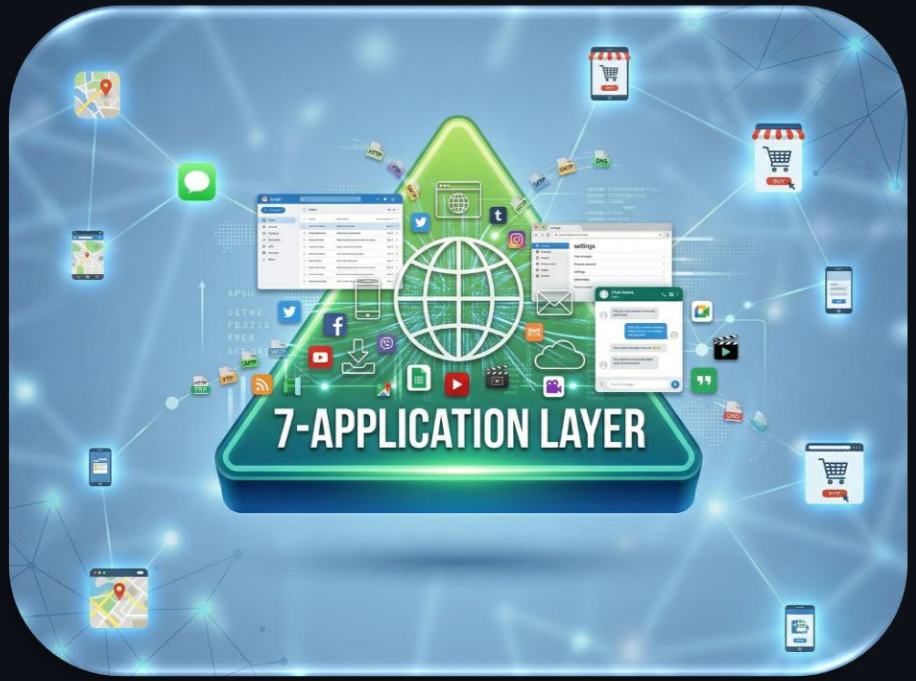
The highest-value target layer.
SQL Injection (SQLi), Cross-Site Scripting (XSS), Command Injection, and all business logic flaws execute here.

PRIMARY THREAT

SQLi / XSS / Logic Flaws

RELEVANT TOOLS

- Burp Suite
- OWASP ZAP
- SQLMap



Key Takeaway

"Map the attack. Secure the layer."

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Full practical video on YouTube

PRACTICE

Map vulnerabilities using OSI

#Cybersecurity #OSIModel #NetworkSecurity #AppSec #LearningInPublic

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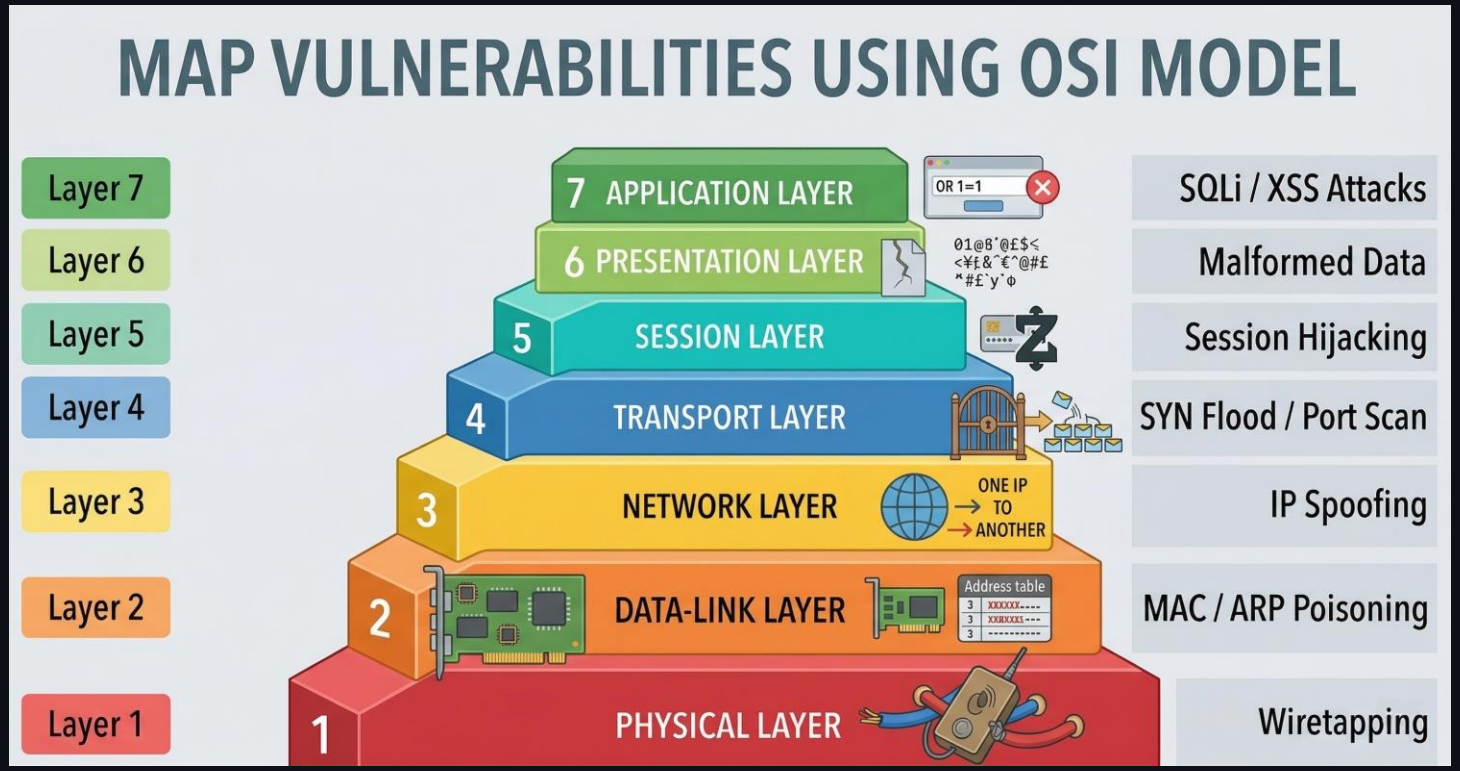
SLIDE 12

PRACTICE

Map vulnerabilities using OSI



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